

2021-Ph-H1-Q15

Section: Particles and Waves

Topic: Radiation

Which graph shows inverse square relationship of radiation intensity with distance?

Solution:

$$I = \frac{k}{r^2}$$

The correct graph **drops steeply at first**, then **levels off** as distance increases.

Answer: C

Guidance for Students:

Inverse square laws give curves that decrease rapidly then flatten out.

Revision Tip:

You must know:

$$I \propto \frac{1}{r^2}$$

for light, radiation, gravity, and electric fields.
